

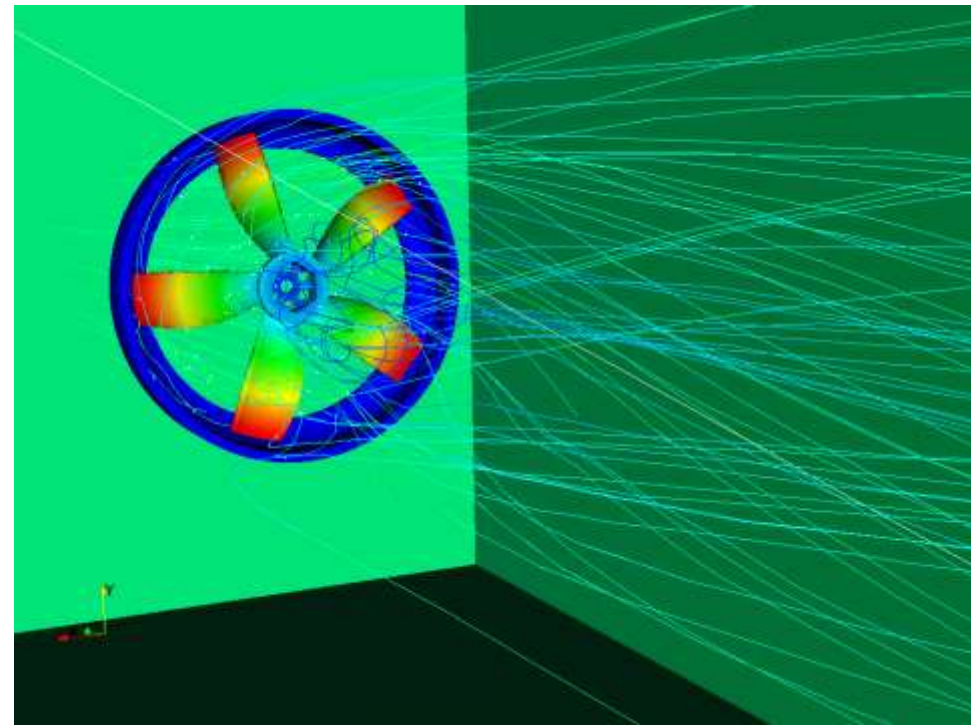
FLUID MECHANICS (ME-205)



AMIT JANGRA
Assistant Professor
Mechanical Engineering Department
DTU

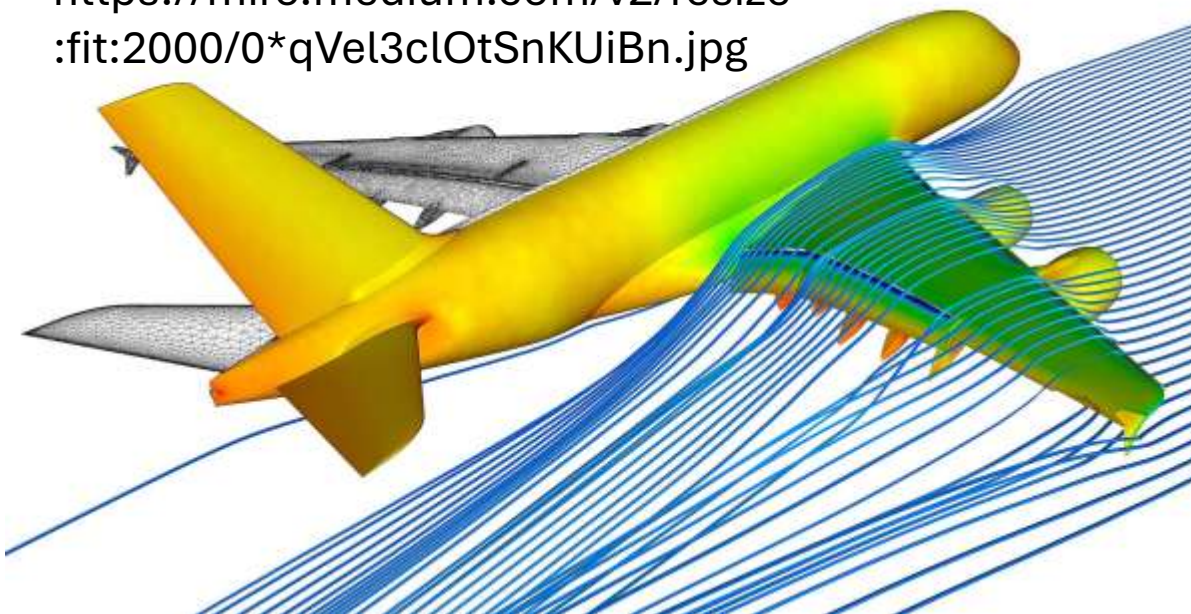


Fluid Mechanics Unit-I



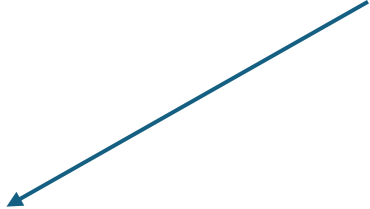
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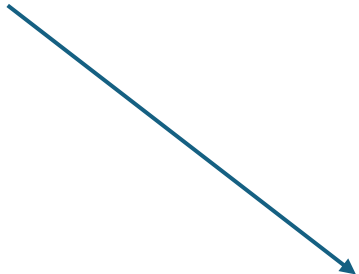


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Fluid Mechanics



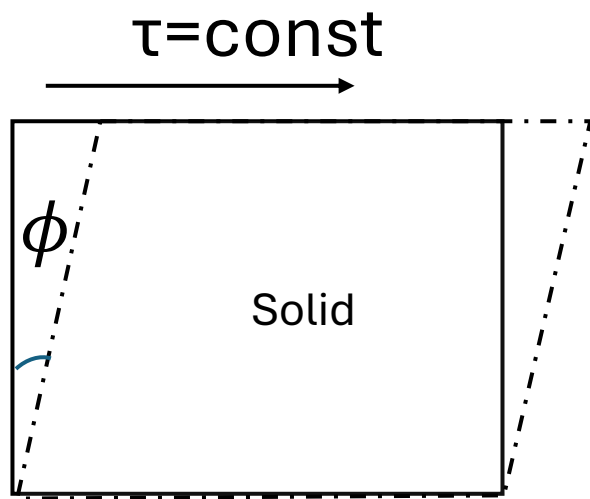
Its existence defines
pressure and shows
continuous deformation
under shear stress Gas or
Liquids



Statics or Dynamics

$$\sum F_{net} = 0 \quad \sum F_{net} > 0$$

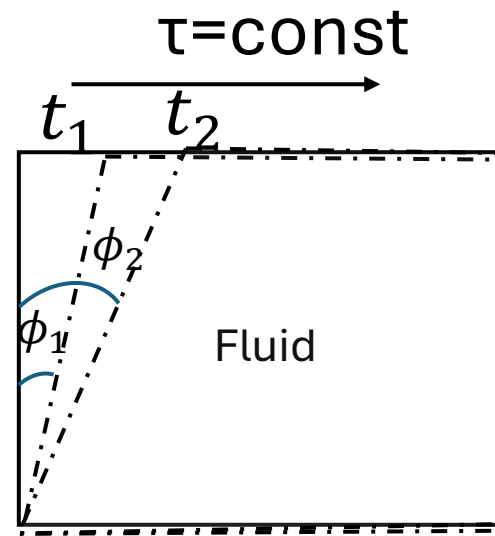
Study of Fluid either in motion (fluid statics) or at rest (fluid dynamics)



$$\tau \propto \phi$$

$$\tau = G\phi$$

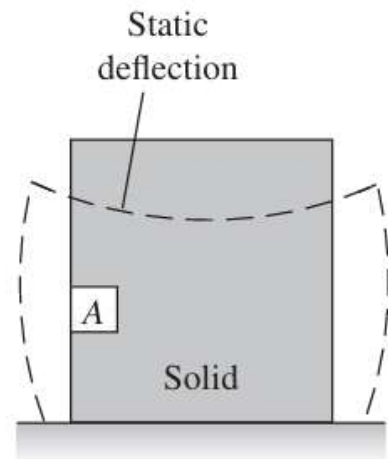
$$G \propto \frac{1}{\phi}$$



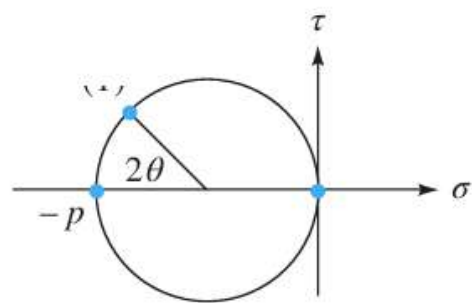
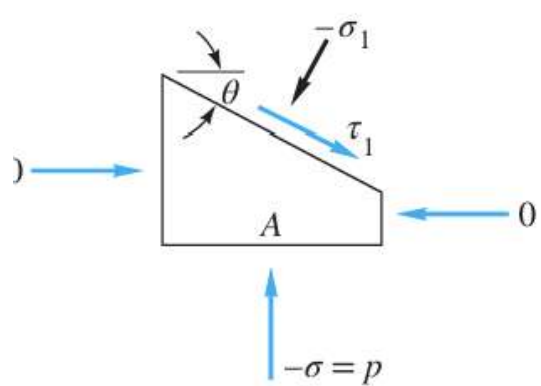
$$\tau \propto \frac{d\phi}{dt}$$

$$\tau = \mu \frac{d\phi}{dt}$$

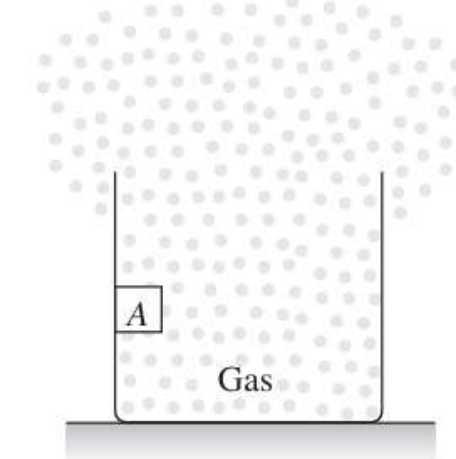
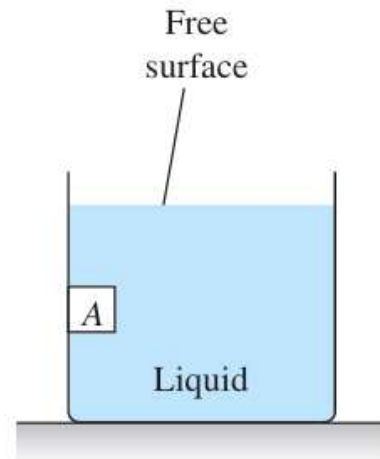
$$\mu = \frac{\tau}{\frac{d\phi}{dt}} \quad \frac{Ns}{m^2}$$



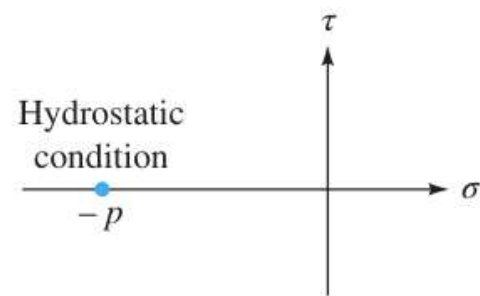
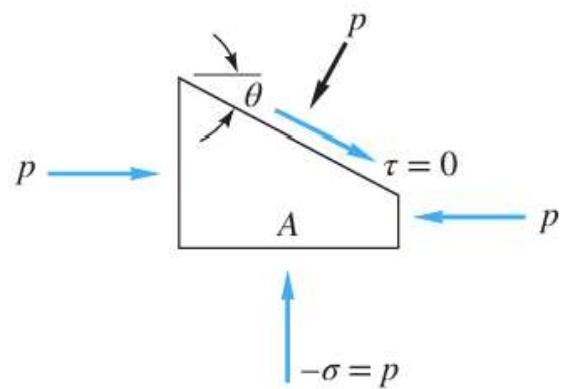
(a)



(b)



(c)



(d)

Common liquids: -

- water,
- oil,
- mercury,
- gasoline,
- alcohol,

Common gases: -

- air,
- helium,
- hydrogen,
- steam,

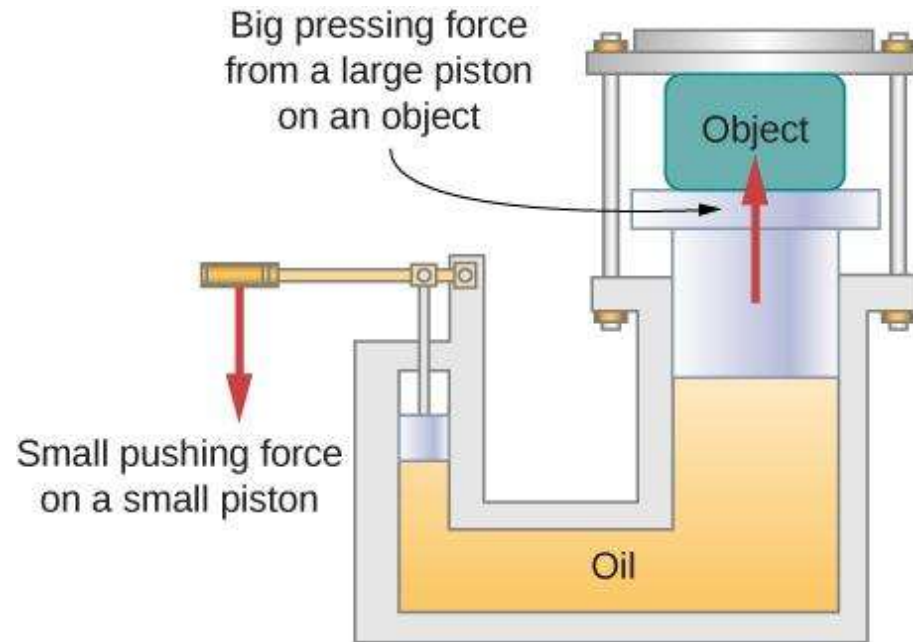
In common temperature and pressure ranges.

Properties of a Fluid

1. Pressure
2. Temperature
3. Density
4. Specific weight
5. Specific gravity
6. Viscosity
7. Bulk modulus
8. Surface tension
9. Vapour pressure

Properties of a Fluid

1. Pressure



Pressure is the **normal force acting per unit area** on a surface. **SI Unit** Pascal (Pa)

$$P = \frac{F}{A}$$

Where P = Pressure (Pa or N/m²); F = Normal force (N); A = Area (m²)

Properties of a Fluid

2. Temperature



Temperature is the degree of hotness or coldness of a body. It indicates the average kinetic energy of the molecules. Units Kelvin (K) (SI); Celsius (°C); Fahrenheit (°F)

$$T(K) = T(^{\circ}C) + 273$$

Water freezes at 0°C (273.15 K) and boils at 100°C (373.15 K) at atmospheric pressure.

Properties of a Fluid

3. Density



Density is the mass per unit volume of a fluid. SI Unit kg/m^3

$$\rho = \frac{m}{V}$$

where m = Mass (kg); V = Volume (m^3)

Typical Values Water = 1000 kg/m^3 ; Air = 1.225 kg/m^3

Properties of a Fluid

4. Specific weight



Specific weight is the **weight per unit volume** of a fluid. **SI Unit** N/m^3 . It indicates **how heavy a fluid is per unit volume**.

$$\gamma = \frac{W}{V} = \rho g$$

where ρ = density; $g = 9.81 \frac{\text{m}}{\text{s}^2}$; W = weight; V = volume

Density tells us **how much mass** is contained in a unit volume.

Specific weight tells us **how much gravitational force (weight)** acts on a unit volume.

Properties of a Fluid

5. Specific gravity

It is the ratio of the density of a fluid to the density of a standard fluid. **Unit** (dimensionless)

$$SG = \frac{\rho_f}{\rho_{std}}$$

Standard fluid for liquids is water, and for gases, air is considered.

$$\rho_{water} = 1000 \frac{kg}{m^3}; \rho_{air} = 1.225 \frac{kg}{m^3}$$

Example

Sp. Gravity of Water = 1; Mercury = 13.6; Kerosene = 0.82

This compare the densities of different fluids and determine whether a fluid is heavier or lighter compared to other fluid.

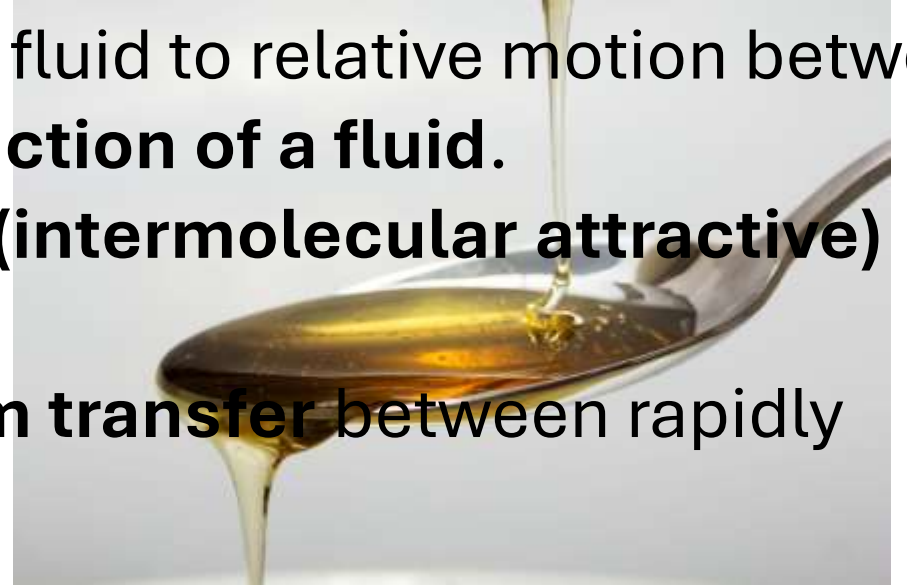
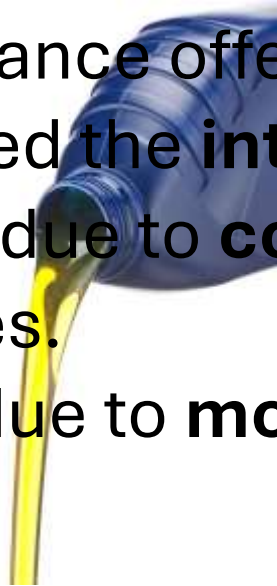
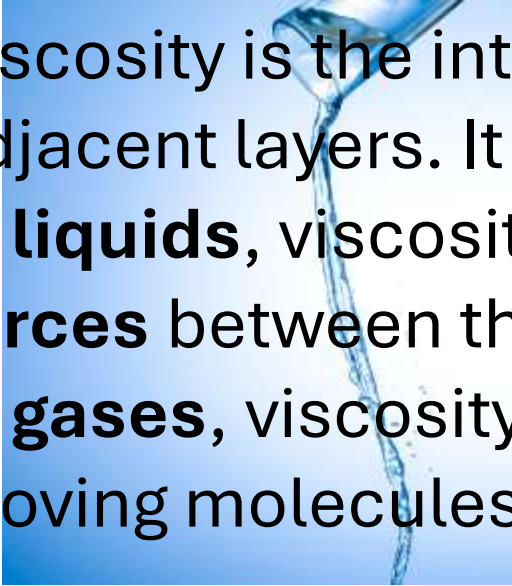
Properties of a Fluid

6. Viscosity

Viscosity is the internal resistance offered by a fluid to relative motion between adjacent layers. It is also called the **internal friction of a fluid**.

In **liquids**, viscosity is mainly due to **cohesive (intermolecular attractive) forces** between the molecules.

In **gases**, viscosity is mainly due to **momentum transfer** between rapidly moving molecules.



Newton's Law of Viscosity.

$$\tau = \mu \frac{du}{dy}$$

where τ = Shear stress; μ = Dynamic viscosity; $\frac{du}{dy}$ = Velocity gradient.

Example: Honey has a much higher viscosity than water

Properties of a Fluid

6. Viscosity

Types

Dynamic viscosity μ (Pa-s)

Kinematic viscosity ν (m²/s)

$$\nu = \frac{\mu}{\rho}$$

Condition

Liquids

Gases

Temperature $\uparrow$

$\mu \downarrow, \nu \downarrow$

$\mu \uparrow, \nu \uparrow$

Pressure $\uparrow$

$\mu \approx \text{constant}, \nu \approx \text{constant}$

$\mu \approx \text{constant}, \nu \downarrow$

Properties of a Fluid

7. Bulk modulus

The bulk modulus measures a fluid's resistance to compression. SI Unit (Pa)

$$K = -v \frac{dP}{dV}$$

Large (K) → Fluid is difficult to compress;

Small (K) → Fluid is easily compressed.

Example: Water has a bulk modulus of approximately 2.2 GPa, making it nearly incompressible.

Properties of a Fluid

8. Surface tension



Surface tension is the property of a liquid by which its free surface behaves like a stretched elastic membrane.

It is caused by the cohesive forces (attractive forces) between liquid molecules.

$$\sigma = \frac{F}{L}$$

Properties of a Fluid

9. Vapour pressure

Vapour pressure is the pressure exerted by the vapour of a liquid when the liquid and its vapour are in thermodynamic equilibrium at a given temperature.

Characteristics

- Increases with temperature.
- A liquid boils when its vapour pressure equals the surrounding atmospheric pressure.

Example

At sea level, water boils when its vapour pressure reaches approximately 101.3 kPa (1 atm).

- Small insects can walk on water.

Properties of a Fluid

9. Vapour Pressure

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ype_id":"VAPOR_PRESSURE"}}?
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Property	Symbol	SI Unit	Formula
Pressure	P	Pa	$(P=\frac{F}{A})$
Temperature	T	K	$(T(K)=T(^{\circ}C)+273.15)$
Density	(ρ)	kg/m^3	$(\rho=\frac{m}{V})$
Specific Weight	(γ)	N/m^3	$(\gamma=\rho g)$
Specific Gravity	SG	Dimensionless	$(SG=\frac{\rho}{\rho_w})$
Dynamic Viscosity	(μ)	Pa.s	$(\tau=\mu \frac{du}{dy})$
Kinematic Viscosity	(ν)	m^2/s	$(\nu=\frac{\mu}{\rho})$
Bulk Modulus	(K)	Pa	$(K=-V \frac{dP}{dV})$
Surface Tension	(σ)	N/m	$(\sigma=\frac{F}{L})$
Vapour Pressure	(P_v)	Pa	Pressure of vapour in equilibrium with liquid

Type of a Fluid

1. Ideal vs Real
2. Newtonian vs Non-Newtonian
3. Compressible vs Incompressible

Type of a Fluid

1. Ideal vs Real

An ideal fluid that has non-viscous or frictionless, incompressible properties. This is an imaginary fluid and does not exist in reality.

All fluids are real and exhibit viscosity, compressibility, and surface tension. E.g. air, water, oil, etc.

Type of a Fluid

Newtonian Fluid

$$\tau = \mu \frac{du}{dy}$$

E.g. water, oil

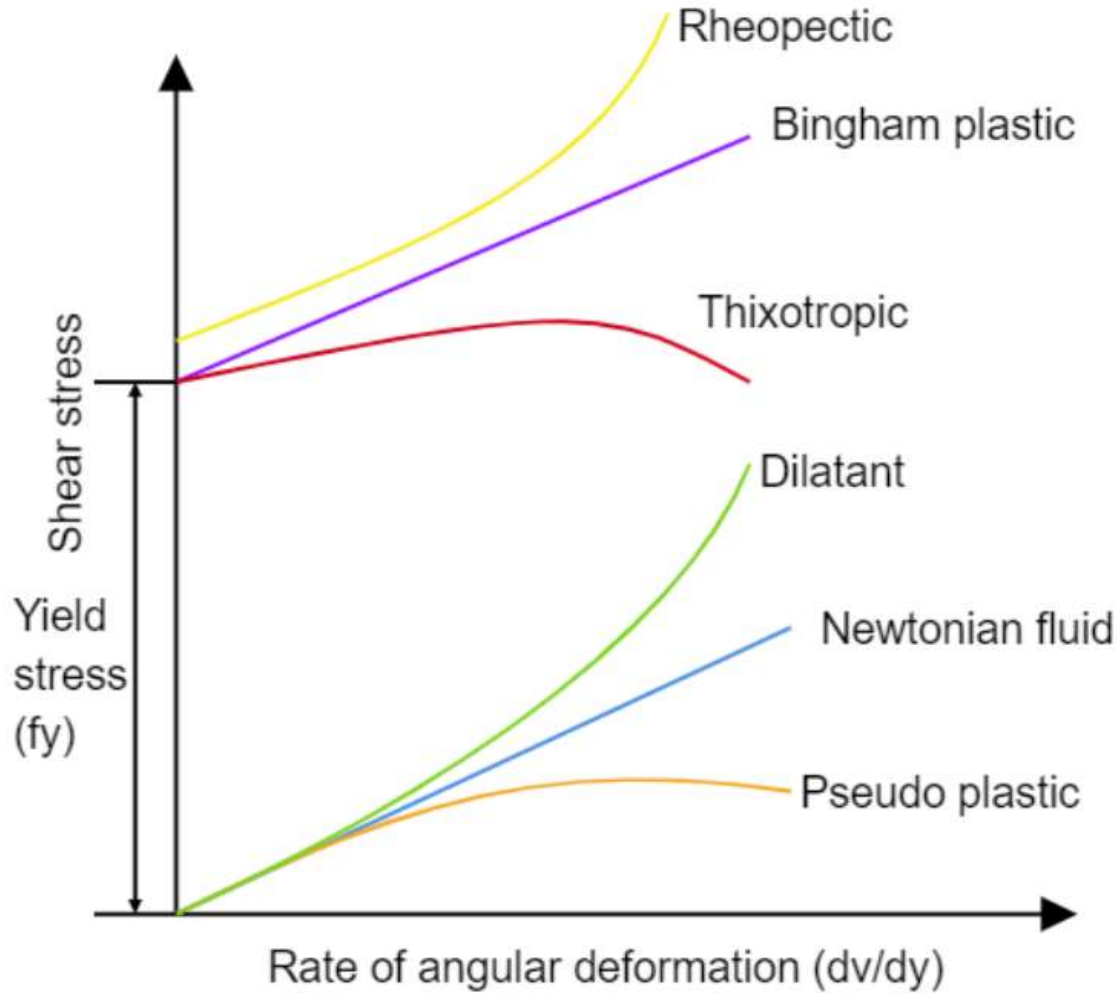
Non-Newtonian Fluid

$$\tau = \mu \left(\frac{du}{dy} \right)^n = \mu \left(\frac{du}{dy} \right)^{n-1} \left(\frac{du}{dy} \right)$$

$$\tau = \mu_{app} \left(\frac{du}{dy} \right)$$

$$\mu_{app} = \mu \left(\frac{du}{dy} \right)^{n-1}$$

Type of a Fluid



1. Pseudoplastic (Shear-Thinning)

Examples: Ketchup, Paint, Blood, Yogurt

2. Dilatant (Shear-Thickening)

Examples Wet sand

3. Bingham Plastic

Examples: Toothpaste, Mayonnaise, Mud, Chocolate paste

4. Thixotropic Fluid

Examples: printing ink, Paint, Clay suspension

5. Rheopectic Fluid

Examples: Gypsum paste, Certain lubricating greases, Some printer inks

Type of a Fluid

Compressible

$$\rho \neq \text{constant}$$

In case of compressible fluid density changes with temperature and pressure

A balloon filled with air
High-speed jet

$$\frac{d\rho}{\rho} < 5\%$$

However, a change in density with in 5% is allowed to consider any fluid to be incompressible.

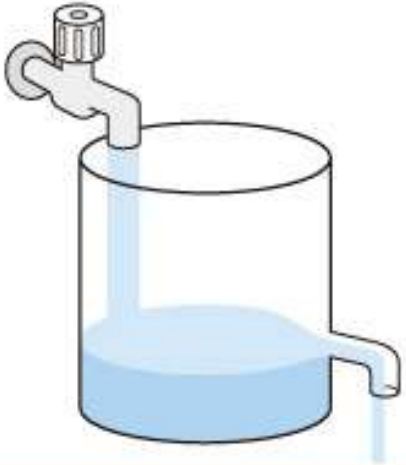
Incompressible

$$\rho = \text{constant}$$

In case of incompressible fluid density does not changes with temperature and pressure

Water

Type of Fluid Flow



Poured water > Discharged water



Poured water = Discharged water

$$\frac{\delta}{\delta x} = 0 \text{ Uniform flow}$$

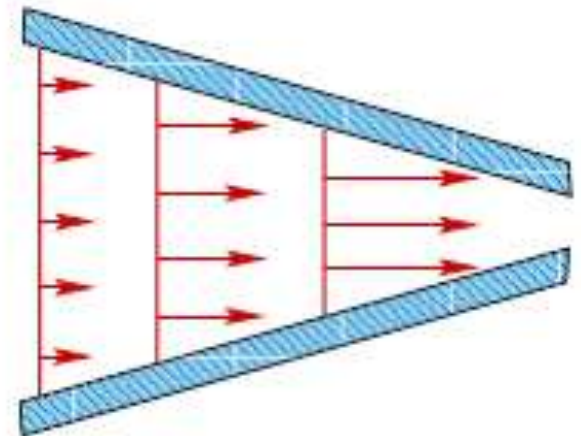
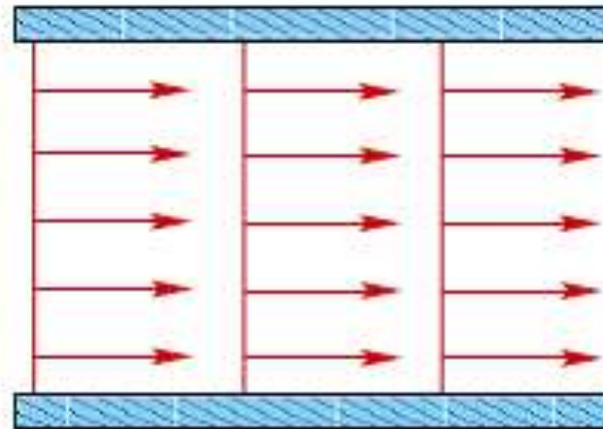
$$\frac{\delta}{\delta x} \neq 0 \text{ Non uniform flow}$$

Uniform and Non-uniform

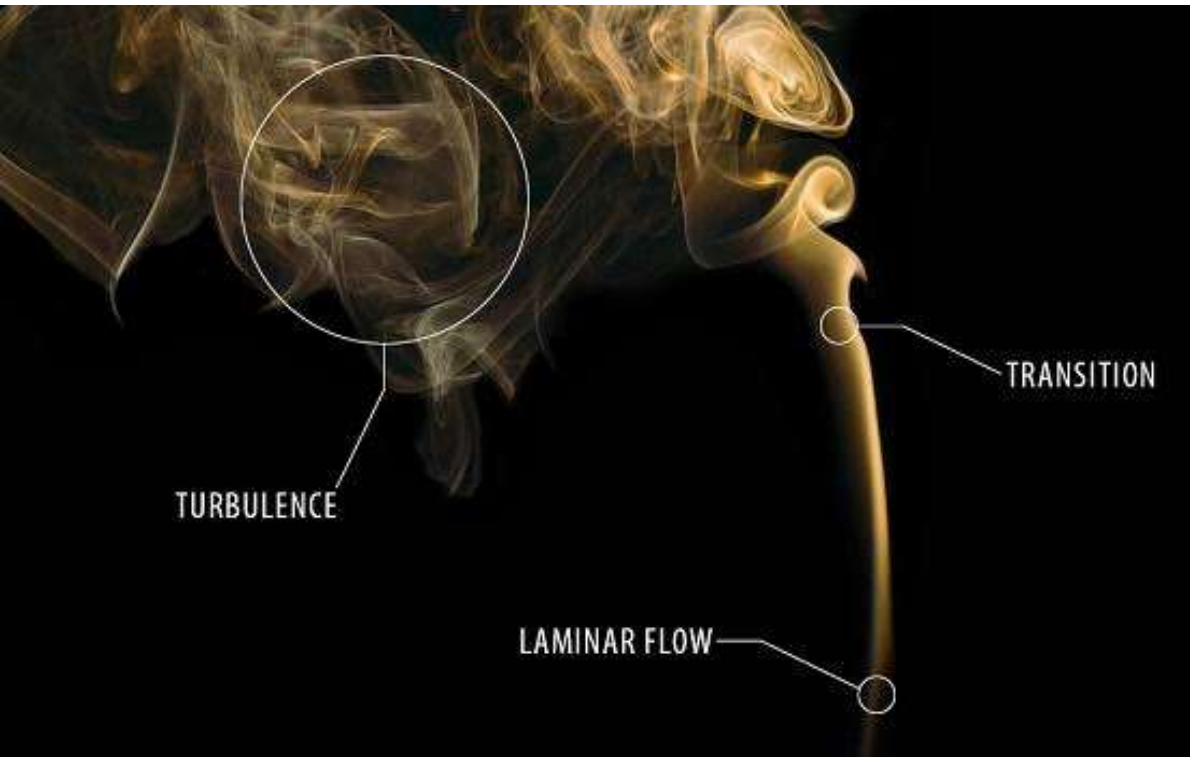
Unsteady and Steady

$$\frac{\delta}{\delta t} \neq 0 \text{ Unsteady flow}$$

$$\frac{\delta}{\delta t} = 0 \text{ Steady Flow}$$



Type of Fluid Flow



Laminar and Turbulent

$$Re = \frac{\rho u_{avg} L_c}{\mu}$$

L_c characteristic dimension

$L_c = d$ for pipe flow

Rotational and Irrotational

